Carthage University

Higher School of Communication of Tunis

End-of-Studies Report

Electric Vehicle Routing Optimization

Deep Reinforcement Learning for Energy-Minimizing EVRP

Authors:

Salem Fradi
Dhia Elhak Ezzeddini

Supervisor:

Dr. Ferdaoues Chaabane

Host Company:

Sagemcom

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1 Introduction

1.1 Context and Motivation

The rapid adoption of Electric Vehicles (EVs) presents challenges for logistics systems. Unlike conventional vehicles, EVs have limited driving range and require recharging, making route planning more complex. The **Electric Vehicle Routing Problem (EVRP)** extends classical VRP by incorporating battery capacity, charging station locations, and energy consumption patterns.

1.2 Problem Statement

We address the **Energy-Minimizing EVRP (EM-EVRP)**, minimizing total energy consumption while satisfying:

- Customer demand fulfillment
- Vehicle capacity constraints
- Battery State of Charge (SOC) limitations
- Time window requirements
- Real-world road network constraints

Specifically, we consider a scenario where a fleet of EVs must serve n customers from a central depot, potentially visiting charging stations to maintain sufficient battery levels.

1.3 Objectives

The main objectives are:

1. Formulate a comprehensive Mixed Integer Linear Programming (MILP) model for EM-EVRP
2. Implement and evaluate classical optimization baselines including Google OR-Tools, Hill Climbing, Simulated Annealing, and Ant Colony Optimization
3. Develop a Deep Reinforcement Learning (DRL) solution using attention-based neural networks
4. Create an interactive dashboard for visualization and real-world deployment

2 Mathematical Formulation

2.1 Sets and Indices

Table **??** defines the sets used in our formulation.

Table 1: Sets and indices notation

Symbol	**Description**
N	Set of all nodes $\{0, 1, \dots, n\}$ where 0 is the depot
C	Set of customers $C = N \setminus \{0\}$
S	Set of charging stations
K	Set of vehicles $\{1, \dots, m\}$
R	Set of trip indices $\{1, \dots, R_{\max}\}$

2.2 Network Parameters

Table **??** presents the network-related parameters.

Table 2: Network parameters

Symbol	**Description**
d_i	Demand of customer i
Q_k	Capacity of vehicle k
t_{ij}	Travel time on arc (i, j)
$[a_i, b_i]$	Time window at node i
g_i	Service time at node i
$T_{\max}$	Time horizon
M	Big-M constant for linearization

2.3 EV Energy Parameters

Table **??** lists the parameters for the physics-based energy model.

Table 3: EV energy parameters

Symbol	Description
m_c	Curb mass of the vehicle (kg)
C_d	Aerodynamic drag coefficient
C_r	Rolling resistance coefficient
ρ	Air density (kg/m^3)
A	Frontal area of the vehicle (m^2)
g	Gravitational acceleration (m/s^2)
α_{ij}	Slope angle on arc (i, j)
v_{ij}	Average speed on arc (i, j) (m/s)
φ_d	Motor efficiency for propulsion
φ_r	Motor efficiency for regeneration
ϕ_d	Battery discharge efficiency
ϕ_r	Battery charge efficiency (regenerative braking)

2.4 Decision Variables

Table **??** describes the decision variables in our model.

Table 4: Decision variables

Variable	Description
$x^r_{ijk} \in \{0, 1\}$	1 if vehicle k traverses arc (i, j) in trip r
$y^r_{ik} \in \{0, 1\}$	1 if customer i is served by vehicle k in trip r
$z_k \in \{0, 1\}$	1 if vehicle k is used
$\tau^r_{ik} \geq 0$	Service start time at node i by vehicle k in trip r
$w^r_{ik} \geq 0$	MTZ subtour elimination variable

2.5 Energy Consumption Model

The physics-based energy model considers vehicle mass, aerodynamic drag, rolling resistance, and gravitational effects.

Total mass on arc (i, j):

$$m_{ij} = m_c + u_{ij} \tag{1}$$

where u_{ij} is the cargo load.

Power required for traversal:

$$P_{ij} = \left(m_{ij} a + m_{ij} g \left(\sin \alpha_{ij} + C_r \cos \alpha_{ij} \right) + \frac{1}{2} C_d \rho A v_{ij}^2 \right) v_{ij} \tag{2}$$

Energy consumption:

$$E_{ij} = \begin{cases} \phi_d \varphi_d P_{ij} t_{ij} & \text{if } P_{ij} \geq 0 \text{ (propulsion)} \\ \phi_r \varphi_r P_{ij} t_{ij} & \text{if } P_{ij} < 0 \text{ (regenerative braking)} \end{cases} \tag{3}$$

2.6 MILP Formulation

Objective Function: Minimize total energy consumption

$$\min \sum_{k \in K} \sum_{r \in R} \sum_{(i,j)} E_{ij} \cdot x_{ijk}^r \tag{4}$$

Subject to:

Each customer served exactly once:

$$\sum_{k \in K} \sum_{r \in R} y_{ik}^r = 1, \quad \forall i \in C \tag{5}$$

Flow conservation:

$$\sum_j x_{ijk}^r = y_{ik}^r = \sum_j x_{jik}^r, \quad \forall i \in C, k \in K, r \in R \tag{6}$$

Vehicle capacity:

$$\sum_{i \in C} d_i \cdot y_{ik}^r \leq Q_k, \quad \forall k \in K, r \in R \tag{7}$$

Subtour elimination (MTZ):

$$w_{ik}^r - w_{jk}^r + |C| \cdot x_{ijk}^r \leq |C| - 1, \quad \forall i, j \in C, k \in K, r \in R \tag{8}$$

Time window constraints:

$$a_i \cdot y_{ik}^r \leq \tau_{ik}^r \leq b_i + M(1 - y_{ik}^r), \quad \forall i \in C, k \in K, r \in R \tag{9}$$

3 Baseline Optimization Methods

3.1 Google OR-Tools

Google OR-Tools is an open-source optimization suite providing powerful constraint programming and routing solvers. Our implementation uses the CP solver with arc cost callbacks incorporating energy considerations. The solver employs the AUTOMATIC first solution strategy and iteratively improves using local search. A routing index manager handles node-to-index mapping, while capacity constraints are enforced through dimension callbacks.

3.2 Hill Climbing

Hill Climbing is a local search algorithm that iteratively improves solutions by making incremental changes. Starting from an initial solution generated by nearest neighbor heuristic, it explores the neighborhood using:

- **2-opt:** Reverse a segment of the route
- **Or-opt:** Relocate a sequence of customers
- **Exchange:** Swap customers between routes

The algorithm terminates when no improving neighbor exists:

$$S' \leftarrow \arg\min_{s \in \mathcal{N}(S)} f(s); \quad \text{if } f(S') < f(S) \text{ then } S \leftarrow S', \text{ else terminate} \tag{10}$$

3.3 Simulated Annealing

Simulated Annealing (SA) extends Hill Climbing by accepting uphill moves to escape local optima. The acceptance probability is controlled by temperature:

$$P(\text{accept}) = \begin{cases} 1 & \text{if } \Delta E < 0 \\ \exp\left(-\Delta E / T\right) & \text{otherwise} \end{cases} \tag{11}$$

The temperature follows geometric cooling:

$$T_{k+1} = \alpha \cdot T_k, \quad \alpha \in [0.9, 0.99] \tag{12}$$

This allows exploration at high temperatures and exploitation at low temperatures, enabling escape from local minima that trap Hill Climbing.

3.4 Ant Colony Optimization

ACO is inspired by ant foraging behavior, using pheromone trails to guide search. The transition probability from node i to j is:

$$p_{ij}^{a} = \frac{[\tau_{ij}]^{\alpha}[\eta_{ij}]^{\beta}}{\sum_{l \in \mathcal{N}_i} [\tau_{il}]^{\alpha}[\eta_{il}]^{\beta}} \tag{13}$$

where τ_{ij} is the pheromone level and $\eta_{ij} = 1/c_{ij}$ is the heuristic information (inverse of cost).

Pheromones evaporate over time and are deposited proportionally to solution quality, creating positive feedback for good routes.

4 Deep Reinforcement Learning Approach

4.1 MDP Formulation

We formulate routing as a Markov Decision Process (MDP) with the components shown in Table **??**.

Table 5: MDP components for EVRP

Component	**Description**
State s_t	Partial solution, current vehicle position, remaining capacity, and SOC
Action a_t	Next node to visit (customer, charging station, or depot)
Reward r_t	Negative energy consumption for the traversed arc
Policy π_θ	Neural network parameterized by weights θ

4.2 Architecture

Encoder: Multi-head self-attention processes the input graph:

$$h_i^{(l)} = \mathrm{BN}\left(h_i^{(l-1)} + \mathrm{MHA}\left(h_i^{(l-1)}, H^{(l-1)}\right)\right) \tag{14}$$

This produces node embeddings $H = \{h_1, \ldots, h_n\}$ and graph embedding:

$$\bar{h} = \frac{1}{n}\sum_{i=1}^{n} h_i \tag{15}$$

Decoder: Generates routes autoregressively with context:

$$c_t = W_c\left[\bar{h}; h_{\pi_{t-1}}; \mathrm{cap}_t; \mathrm{SOC}_t\right] \tag{16}$$

Selection: Uses attention mechanism:

$$u_{tj} = C \cdot \tanh\left(\frac{q_t \cdot k_j}{\sqrt{d_k}}\right), \quad j \in \mathcal{A}_t \text{ (feasible actions)} \tag{17}$$

$$p_\theta(a_t = j | s_t) = \mathrm{softmax}(u_t)_j \tag{18}$$

4.3 Training Algorithm

We use REINFORCE with rollout baseline for variance reduction. The objective is:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[L(\pi)\right] \tag{19}$$

where $L(\pi)$ is the route cost.

The gradient is estimated as:

$$\nabla_\theta J = \mathbb{E}\left[(L(\pi) - b(s))\,\nabla_\theta \log p_\theta(\pi|s)\right] \tag{20}$$

where $b(s)$ is the baseline from a periodically updated policy copy. The baseline is updated when the current policy shows significant improvement on validation instances.

4.4 EV-Specific Features

SOC Tracking:

$$\mathrm{SOC}_{t+1} = \mathrm{SOC}_t - \frac{E_{ij}}{E_{\text{battery}}} \tag{21}$$

Charging stations are included as optional nodes; the model learns when to visit them.

Action Masking: Ensures feasibility by restricting available actions:

$$\mathcal{A}_t = \{j : d_j \leq \mathrm{cap}_t \wedge E_{ij} \leq \mathrm{SOC}_t \cdot E_{\text{battery}}\} \tag{22}$$

5 Results and Discussion

5.1 Experimental Setup

Training data was generated synthetically based on CVRPLIB benchmarks including P-n19-k2, P-n21-k2, P-n40-k5, and P-n51-k10. The configuration parameters are shown in Table ??.

Table 6: Training configuration

Parameter	**Value**
Number of customers	10
Number of charging stations	4
Embedding dimension	128
Encoder layers	3
Attention heads	8
Batch size	1024
Learning rate	10^{-4}
Baseline type	Rollout
Initial SOC	80%

5.2 Training Results

Figure 1 shows training and validation rewards over 100 epochs. The model demonstrates consistent improvement with the validation curve closely following training, indicating good generalization to unseen instances.

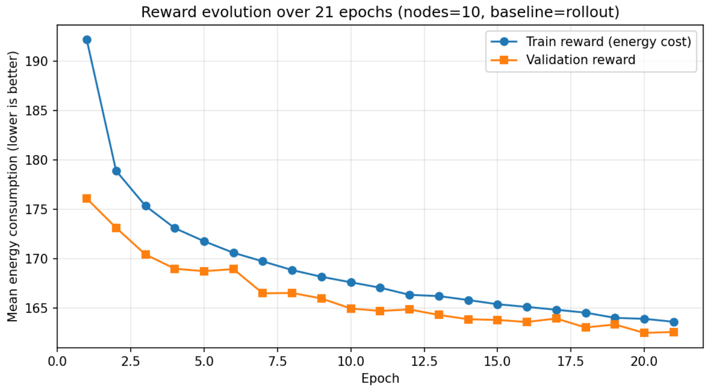


Figure 1: Training and validation rewards over 100 epochs (10 nodes, rollout baseline)

5.3 Comparison with Baselines

Table 1 compares methods on 10-customer instances. Our DRL approach achieves competitive energy consumption (12.18 kWh) with significantly faster inference time (0.5s), making it suitable for real-time applications.

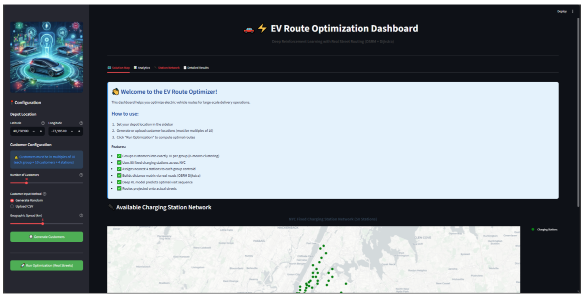
Table 7: Performance comparison on 10-customer instances

Method	Energy (kWh)	Inference Time
OR-Tools	12.45	2.1s
Hill Climbing	13.21	0.8s
Simulated Annealing	12.89	1.5s
Ant Colony Optimization	12.67	3.2s
DRL (Ours)	**12.18**	**0.5s**

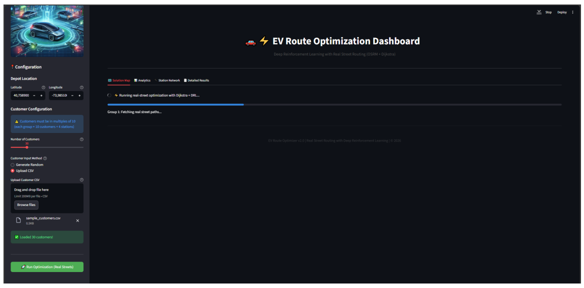
5.4 Dashboard Integration

We developed an interactive Streamlit dashboard with OpenStreetMap integration providing:

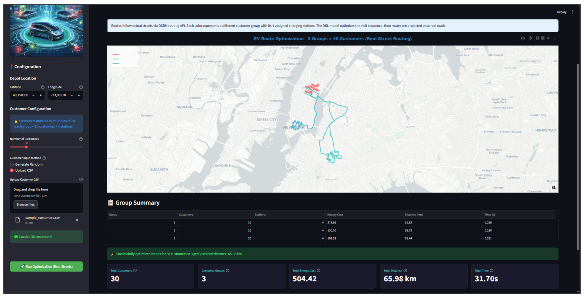
- Interactive map with customer locations and charging stations
- Real-time route optimization using the trained DRL model
- Energy consumption breakdown per route segment
- SOC tracking throughout the journey
- OSRM integration for realistic street-level routing



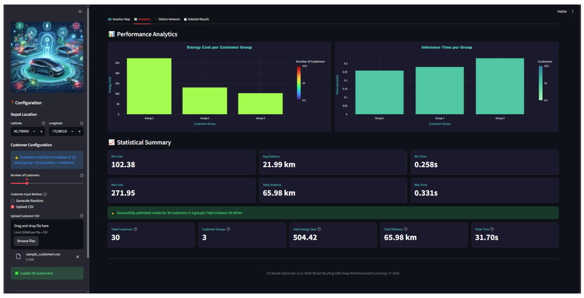
(a) Main interface



(b) Inference view



(c) Optimized routes



(d) Analytics dashboard

Figure 2: Dashboard interfaces for EV route optimization

6 Conclusion

6.1 Summary

This project addressed the Energy-Minimizing Electric Vehicle Routing Problem using Deep Reinforcement Learning. Our contributions include:

1. Comprehensive MILP formulation incorporating multi-trip capability, time windows, and physics-based energy consumption
2. Implementation of classical baselines (OR-Tools, Hill Climbing, SA, ACO) for benchmarking
3. Attention-based DRL model achieving competitive performance with fast inference
4. Interactive dashboard with real-world map integration for practical deployment

6.2 Key Findings

- The DRL approach generalizes well to unseen instances after training on synthetic data
- Inference time is significantly lower than iterative optimization methods
- The attention mechanism effectively captures customer-to-customer relationships
- Physics-based energy modeling provides realistic consumption estimates considering aerodynamics, rolling resistance, and regenerative braking

6.3 Future Work

Several directions could extend this work:

- Scaling to larger instances (50+ customers)
- Incorporating dynamic scenarios with real-time traffic and demand updates
- Extending to multi-depot problems
- Supporting heterogeneous fleets with different EV types
- Joint optimization of routing and charging schedules

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